

Adding Landscape Explanatory Variables to ‘SSN’ Objects Directly in R Using StreamCatTools

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Background

Landscape explanatory variables can be added directly to SSN objects using the **StreamCatTools** R package. **StreamCatTools** interacts with StreamCat (Hill et al. 2016), a database of hundreds of metrics for approximately 2.65 million stream segments in the conterminous United States (CONUS). StreamCat metrics exist at catchment and watershed scales and represent natural and anthropogenic landscape variables. StreamCat is built on the medium resolution National Hydrography Dataset Plus (version 2.1). To learn more about StreamCat, visit their website at [this link](#). In this vignette, we will show how to use **StreamCatTools** to access StreamCat data; for more detail regarding **StreamCatTools**, visit their website at [this link](#).

Before proceeding, we load the **SSN2**, **StreamCatTools**, and **ggplot2** R packages (which can all be installed directly from CRAN) by running:

```
library(SSN2)
library(StreamCatTools)
library(ggplot2)
```

The bugs Data

The **bugs** data is an SSN object that contains macroinvertebrate data in the Lower Snake Basin. It can be downloaded via GitHub at [this link](#). We read in **bugs** by running:

```
bugs <- ssn_import("bugs.ssn", predpts = "pred")
```

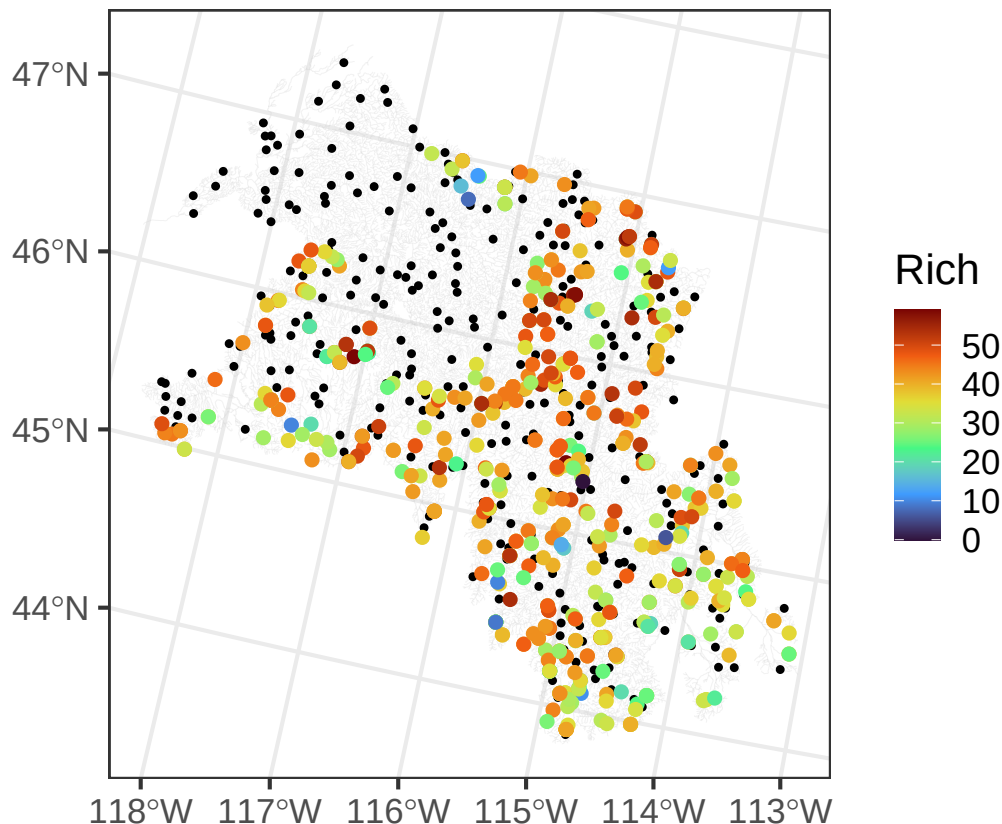
There are 549 observed sites and 300 prediction sites:

```
bugs
#> Object of class SSN
#>
#> Object includes observations on 13 variables across 549 sites within the bounding box
#>      xmin      ymin      xmax      ymax
#> -1765346 2459843 -1343175 2909567
#>
#> Object also includes 1 set of prediction points with 300 locations
#>
#> Variable names are (found using ssn_names(object)):
#> $obs
#> [1] "comid"      "ELEV_DEM"   "AREAWTMAP"  "Rich"       "rid"        "ratio"
#> [7] "upDist"     "afvArea"    "locID"      "netID"      "pid"        "geometry"
#> [13] "netgeom"
#>
#> $pred
#> [1] "comid"      "ELEV_DEM"   "AREAWTMAP"  "rid"        "ratio"      "upDist"
```

```
#> [7] "afvArea" "locID" "netID" "pid" "geom" "netgeom"
```

We visualize the flowlines (edges), observed macroinvertebrate richness (obs), and prediction sites (preds) by running:

```
ggplot() +
  geom_sf(data = bugs$edges, linewidth = 0.1, alpha = 0.3, color = "lightgrey") +
  geom_sf(data = bugs$preds$pred, color = "black", size = 0.9) +
  geom_sf(data = bugs$obs, aes(color = Rich), size = 2) +
  scale_color_viridis_c(option = "H", limits = c(0, 59)) +
  theme_bw(base_size = 16)
```



We create the hydrologic distance matrices required for fitting SSN models by running:

```
ssn_create_distmat(bugs, predpts = "pred")
```

Adding Explanatory Variables to an SSN Object

Any of the hundreds of StreamCat metrics can be added to observed and prediction sites through NHD's common identifier, or COMID. The COMID is a 10 digit code that uniquely identifies an NHD feature. The `bugs` data has COMIDs available for each observed and prediction site. StreamCat metrics can be searched at [this link](#) or within `StreamCatTools`, which we discuss later.

In StreamCat, the `"tmean8110"` metric contains 30-year mean annual temperature between 1981 and 2010, and the `"pctconif2008"` contains the proportion of evergreen forest land cover in 2008 (based on the National Landcover Dataset, or NLCD; see [this link](#) for more). StreamCat metrics are defined with respect to an "Area of Interest", or AOI. Typically these are "cat" (for local catchment) or "ws" (for full upstream

watershed). Additional AOIs exist for catchments and watersheds restricted to 100 meter riparian buffer. The `sc_get_data()` function in `StreamCatTools` returns metrics in the AOI based on COMID:

```
sc_vars <- c("tmean8110", "pctconif2008")
sc_data <- sc_get_data(
  metric = sc_vars,
  aoi = "cat",
  comid = bugs$obs$comid
)
head(sc_data)
#>      comid pctconif2008cat tmean8110cat
#> 1 23428382          85.10      7.663872
#> 2 23428446          93.14      6.605735
#> 3 23428578          94.49      6.685206
#> 4 23428848          94.16      6.353155
#> 5 23428858          90.34      6.584575
#> 6 23428892          99.89      4.959707
```

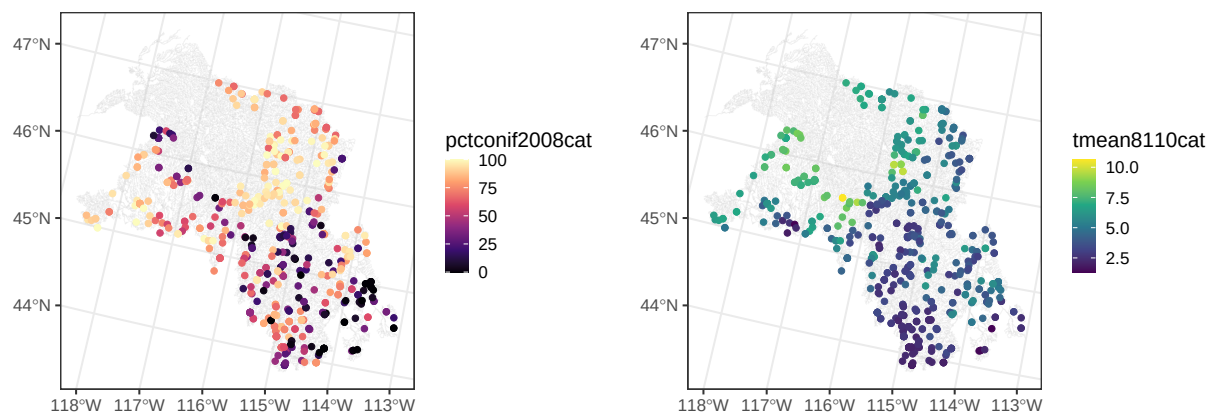
`sc_data` is a data frame with column `comid`, column `pctconif2008cat` (for `pctconif2008` in the catchment AOI), and column `tmean8110cat` (for `tmean8110cat`). These metrics are merged to `bugs$obs` via "`comid`" by running:

```
bugs$obs <- merge(bugs$obs, sc_data, by = "comid")
```

We visualize these new metrics spatially by running:

```
ggplot() +
  geom_sf(data = bugs$edges, linewidth = 0.1, alpha = 0.3, color = "lightgrey") +
  geom_sf(data = bugs$obs, aes(color = pctconif2008cat), size = 2) +
  scale_color_viridis_c(option = "A") +
  theme_bw(base_size = 16)

ggplot() +
  geom_sf(data = bugs$edges, linewidth = 0.1, alpha = 0.3, color = "lightgrey") +
  geom_sf(data = bugs$obs, aes(color = tmean8110cat), size = 2) +
  scale_color_viridis_c(option = "D") +
  theme_bw(base_size = 16)
```



Similarly, these metrics can be added to the prediction sites:

```
sc_pred_data <- sc_get_data(
  metric = sc_vars,
  aoi = "cat",
```

```

  comid = bugs$preds$pred$comid
)
bugs$preds$pred <- merge(bugs$preds$pred, sc_pred_data, by = "comid")

```

Fitting an SSN Model

We fit an SSN model describing species richness (the number of unique species) as a function of site elevation, 30-year mean annual temperature, and the percentage of the catchment with evergreen forest cover by running:

```

ssn_mod <- ssn_lm(
  formula = Rich ~ ELEV_DEM + tmean8110cat + pctconif2008cat,
  ssn.object = bugs,
  tailup_type = "exponential",
  taildown_type = "exponential",
  additive = "afvArea"
)
summary(ssn_mod)
#>
#> Call:
#> ssn_lm(formula = Rich ~ ELEV_DEM + tmean8110cat + pctconif2008cat,
#>       ssn.object = bugs, tailup_type = "exponential", taildown_type = "exponential",
#>       additive = "afvArea")
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -34.939  -4.916   2.135   7.168  23.143
#>
#> Coefficients (fixed):
#>              Estimate Std. Error z value Pr(>|z|)
#> (Intercept)    58.222785   8.610861   6.762 1.37e-11 ***
#> ELEV_DEM       -0.010409   0.003305  -3.150  0.00163 **
#> tmean8110cat   -1.442592   0.760347  -1.897  0.05779 .
#> pctconif2008cat 0.027568   0.016689   1.652  0.09857 .
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Pseudo R-squared: 0.03298
#>
#> Coefficients (covariance):
#>              Effect      Parameter      Estimate
#>   tailup exponential de (parsill)         2.97
#>   tailup exponential range      593762.56
#>   taildown exponential de (parsill)        25.82
#>   taildown exponential range      47730.45
#>              nugget      nugget        60.30

```

Here, we modeled the spatial covariance using both a tailup exponential structure and a taildown exponential structure (we omitted a Euclidean structure). Based on the fitted model, there is strong evidence that increasing elevation is associated decreasing richness, on average (p -value < 0.01), some evidence that increasing temperature is associated with decreasing richness, on average (p -value ≈ 0.06), and some evidence that increasing percent evergreen forest cover is associated with increasing richness, on average (p -value ≈ 0.10).

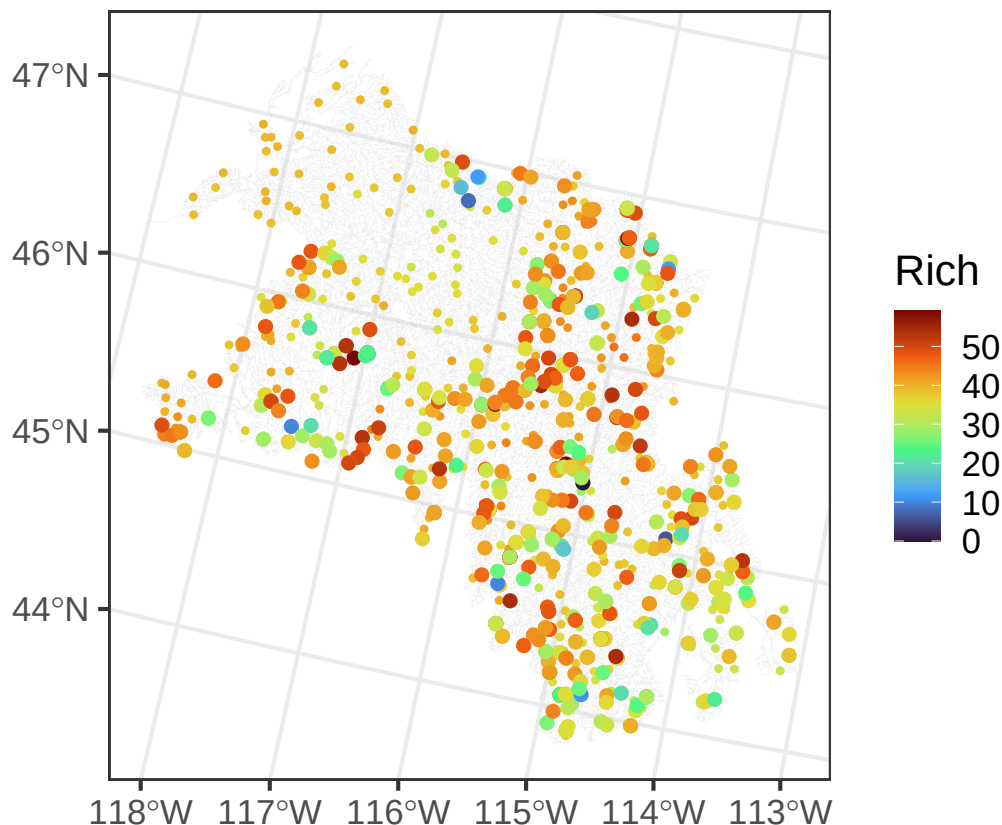
Prediction (Kriging)

We augment the prediction sites with predictions by running:

```
aug_preds <- augment(ssn_mod, newdata = "pred")
```

We visualize these predictions alongside the observed richness by running:

```
ggplot() +  
  geom_sf(data = bugs$edges, linewidth = 0.1, alpha = 0.3, color = "lightgrey") +  
  geom_sf(data = aug_preds, aes(color = .fitted), size = 0.9) +  
  geom_sf(data = bugs$obs, aes(color = Rich), size = 2) +  
  scale_color_viridis_c(option = "H", name = "Rich", limits = c(0, 59)) +  
  theme_bw(base_size = 16)
```



We use block Kriging to predict the average species richness in the Lower Snake Basin alongside a prediction interval by running:

```
predict(ssn_mod, newdata = "pred", block = TRUE, interval = "prediction")  
#>      fit      lwr      upr  
#> 1 37.64725 36.0336 39.26091
```

Other StreamCatTools Features

There are several other **StreamCatTools** features which simplify the process of interacting with StreamCat. The `sc_get_params()` function returns parameters available using the StreamCat API. For example, we can learn details like AOI, category, units, and other metadata information for each metric by running:

```

var_info <- sc_get_params(param = "variable_info")
head(var_info)
#> # A tibble: 6 x 13
#>   category      metric aoi   year short_description long_description units  dsid
#>   <chr>         <chr> <chr> <chr> <chr>               <chr>         <chr> <dbl>
#> 1 Anthropogen~ NABD_~ Cat,~ <NA> NABD Dam Density Density of geor~ Coun~ 33
#> 2 Anthropogen~ NABD_~ Cat,~ <NA> NABD NID Reservo~ Volume all rese~ Cubi~ 33
#> 3 Anthropogen~ NABD_~ Cat,~ <NA> NABD Normal Rese~ Volume all rese~ Cubi~ 33
#> 4 Natural      Preci~ Cat,~ <NA> Surplus Precipit~ This dataset re~ Kilo~ 75
#> 5 Soils         Rckde~ Cat,~ <NA> Mean Bedrock Dep~ Mean depth (cm)~ Cent~ 56
#> 6 Soils         agkff~ Cat,~ <NA> Ag Soil Erodibil~ Mean of STATSGO~ Unit~ 28
#> # i 5 more variables: dataset <chr>, source_name <chr>, source_URL <chr>,
#> # ACTIVE <dbl>, DSNAME <chr>

```

You can search for specific metrics that satisfy certain conditions using `sc_get_metric_names()`. For example, we can find all the climate variables measured at catchment and watershed scales by running:

```

sc_get_metric_names(category = "Climate", aoi = c("Cat", "Ws"))
#> # A tibble: 11 x 9
#>   Category Metric      AOI   Year Short_Name Metric_Description Units Source
#>   <chr>    <chr>    <chr> <chr> <chr>      <chr>         <chr> <chr>
#> 1 Climate bfi[AOI] Cat,~ <NA> Base Flow~ Base flow is the ~ Perc~ USGS ~
#> 2 Climate precip8110[A~ Cat,~ <NA> 30-year M~ PRISM climate dat~ Mill~ PRISM
#> 3 Climate precip9120[A~ Cat,~ 1991~ 30-year M~ PRISM climate dat~ Mill~ PRISM
#> 4 Climate precip[Year]~ Cat,~ 2008~ Mean Prec~ PRISM climate dat~ Mill~ PRISM
#> 5 Climate tmax8110[AOI] Cat,~ 1981~ 30-year A~ PRISM climate dat~ Cels~ PRISM
#> 6 Climate tmax9120[AOI] Cat,~ 1991~ 30-year A~ PRISM climate dat~ Cels~ PRISM
#> 7 Climate tmean8110[AO~ Cat,~ 1981~ 30-year M~ PRISM climate dat~ Cels~ PRISM
#> 8 Climate tmean9120[AO~ Cat,~ 1991~ 30-year M~ PRISM climate dat~ Cels~ PRISM
#> 9 Climate tmean[Year][~ Cat,~ 2008~ Mean Air ~ PRISM climate dat~ Cels~ PRISM
#> 10 Climate tmin8110[AOI] Cat,~ 1981~ 30-year A~ PRISM climate dat~ Cels~ PRISM
#> 11 Climate tmin9120[AOI] Cat,~ 1991~ 30-year A~ PRISM climate dat~ Cels~ PRISM
#> # i 1 more variable: Dataset <chr>

```

If we only have access to x- and y-coordinates (instead of COMIDs), we can use `sc_get_comid()` to find the NHD feature that contains a coordinate by running:

```
sct_comid <- sc_get_comid(bugs$obs)
```

The COMIDs are stored in a single comma-separated vector (this is how StreamCat interacts its API). This vector can be separated into distinct COMIDs for more direct use with **R** by running

```
sct_comid <- unlist(strsplit(sct_comid, ","))
```

The COMIDs stored in `bugs$obs` match the COMIDs we accessed via `sc_get_comid()`:

```

dat_comid <- data.frame(
  bugs_comid = bugs$obs$comid,
  sct_comid = sct_comid
)
head(dat_comid)
#>   bugs_comid sct_comid
#> 1   23428382 23428382
#> 2   23428382 23428382
#> 3   23428446 23428446
#> 4   23428578 23428578

```

```
#> 5    23428848  23428848
#> 6    23428858  23428858
```

Here, we have only briefly explored the potential for SSN2 integration with `StreamCatTools`; to learn more about `StreamCatTools`, visit their website at [this link](#).

Alternatives to StreamCat

Several other landscape data sources exist for stream networks outside of CONUS. In Canada, there is the Canadian Hydrospace Network ([this link](#)). In Europe, there is EU-Hydro ([this link](#)). In Australia, there is Geofabric ([this link](#)). Worldwide there is HydroAtlas ([this link](#)), though HydroAtlas data tends to be coarser than more localized data sets.

R Code Appendix

```
knitr::opts_chunk$set(
  collapse = TRUE,
  comment = "#>",
  message = FALSE,
  warning = FALSE
)
library(SSN2)
library(StreamCatTools)
library(ggplot2)
bugs <- ssn_import("bugs.ssn", predpts = "pred")
bugs
ggplot() +
  geom_sf(data = bugs$edges, linewidth = 0.1, alpha = 0.3, color = "lightgrey") +
  geom_sf(data = bugs$preds$pred, color = "black", size = 0.9) +
  geom_sf(data = bugs$obs, aes(color = Rich), size = 2) +
  scale_color_viridis_c(option = "H", limits = c(0, 59)) +
  theme_bw(base_size = 16)
ssn_create_distmat(bugs, predpts = "pred")
sc_vars <- c("tmean8110", "pctconif2008")
sc_data <- sc_get_data(
  metric = sc_vars,
  aoi = "cat",
  comid = bugs$obs$comid
)
head(sc_data)
bugs$obs <- merge(bugs$obs, sc_data, by = "comid")
ggplot() +
  geom_sf(data = bugs$edges, linewidth = 0.1, alpha = 0.3, color = "lightgrey") +
  geom_sf(data = bugs$obs, aes(color = pctconif2008cat), size = 2) +
  scale_color_viridis_c(option = "A") +
  theme_bw(base_size = 16)

ggplot() +
  geom_sf(data = bugs$edges, linewidth = 0.1, alpha = 0.3, color = "lightgrey") +
  geom_sf(data = bugs$obs, aes(color = tmean8110cat), size = 2) +
  scale_color_viridis_c(option = "D") +
  theme_bw(base_size = 16)
```

```

sc_pred_data <- sc_get_data(
  metric = sc_vars,
  aoi = "cat",
  comid = bugs$preds$pred$comid
)
bugs$preds$pred <- merge(bugs$preds$pred, sc_pred_data, by = "comid")
ssn_mod <- ssn_lm(
  formula = Rich ~ ELEV_DEM + tmean8110cat + pctconif2008cat,
  ssn.object = bugs,
  tailup_type = "exponential",
  taildown_type = "exponential",
  additive = "afvArea"
)
summary(ssn_mod)
aug_preds <- augment(ssn_mod, newdata = "pred")
ggplot() +
  geom_sf(data = bugs$edges, linewidth = 0.1, alpha = 0.3, color = "lightgrey") +
  geom_sf(data = aug_preds, aes(color = .fitted), size = 0.9) +
  geom_sf(data = bugs$obs, aes(color = Rich), size = 2) +
  scale_color_viridis_c(option = "H", name = "Rich", limits = c(0, 59)) +
  theme_bw(base_size = 16)
predict(ssn_mod, newdata = "pred", block = TRUE, interval = "prediction")
var_info <- sc_get_params(param = "variable_info")
head(var_info)
sc_get_metric_names(category = "Climate", aoi = c("Cat", "Ws"))
sct_comid <- sc_get_comid(bugs$obs)
sct_comid <- unlist(strsplit(sct_comid, ","))
dat_comid <- data.frame(
  bugs_comid = bugs$obs$comid,
  sct_comid = sct_comid
)
head(dat_comid)

```

References

Hill, Ryan A, Marc H Weber, Scott G Leibowitz, Anthony R Olsen, and Darren J Thornbrugh. 2016. "The Stream-Catchment (StreamCat) Dataset: A Database of Watershed Metrics for the Conterminous United States." *JAWRA Journal of the American Water Resources Association* 52 (1): 120–28.